

Project Report: Bitcoin Price Forecasting

Project Overview:

This project aims to forecast Bitcoin prices using historical data extracted from Yahoo Finance. The analysis involves two main phases:

1. **Phase 1:** Web scraping and data extraction
2. **Phase 2:** Time series forecasting using the ARIMA model and data visualization

Phase 1: Data Collection via Web scraping

Objective:

To collect historical daily Bitcoin price data and store it in a structured CSV file for later analysis and modeling.

Tools and Methods:

- Libraries used: requests, BeautifulSoup, pandas
- Sent a GET request to Yahoo Finance with browser like headers.
- Parsed the HTML table containing historical prices using BeautifulSoup.
- Extracted and cleaned the "Date" and "Close Price" columns
- Saved the data as a CSV file name btc_prices.csv

Phase 2: Time Series Forecasting with ARIMA

Objective:

To forecast Bitcoin's closing price for the next 20 days using ARIMA model and visualize the result.

Steps and Methods

1. Data Preparation:

- Loaded btc_data.csv using pandas
- Cleaned columns and converted Date to datetime
- Reversed data chronological order

2. Model Training:

- Fitted an ARIMA model with parameters (2, 1, 10) on closing price

3. Forecasting:

- Forecasted the next 20 days prices
- Created future dates starting from the last date in the dataset
- Saved forecasted data in btc_prices_predicted.csv

4. Visualization:

- Plotted actual prices in blue and forecasted prices in green using matplotlib
- Saved the plot as plot.png

Outputs

- Btc_prices-preficted.csv : CSV file containing forecast dates and closing prices for 20 days
- Plot.png : Visualization comparing actual and predicted prices

Conclusion

This project successfully demonstrates:

- How to extracted real world financial data via web scraping
- How to apply time series forecasting with ARIMA
- How to save and visualize result effectively

This foundational approach to Bitcoin price prediction can be extended with more advance models in future work.